

Survey Report on Emerging Trends in Low-Code Application Development

Tejini V
241401115,CSBS

Popular Low-Code Development Platforms

1. Microsoft Power Apps

Introduction

Microsoft Power Apps is a low-code application development platform developed by Microsoft. It allows users to create business applications using visual interfaces and minimal manual coding. It is part of the Microsoft Power Platform and works closely with Microsoft 365 and Azure services. The platform is mainly used to develop business applications, automate processes, and connect organizational data.

Developer

- **Developer:** Microsoft Corporation.
- **Platform:** Microsoft Power Platform.
- **Primary focus:** Rapid development of business applications.
- **Target users:** Business users, citizen developers, and professional developers.

Key Features

- **Visual development:** Applications can be created using drag-and-drop controls and low-code configuration.
- **Multiple app types:** Supports Canvas Apps and Model-Driven Apps for different development requirements.
- **Data integration:** Provides integration with Dataverse, SharePoint, SQL Server, Excel, and many other services.
- **AI and automation:** Integrates AI assistance and Power Automate for intelligent workflows.

Advantages

- **Fast development:** Applications can be created much faster than through conventional coding.
- **Easy integration:** Works particularly well with Microsoft 365, Dynamics 365, Azure, and Dataverse.
- **Low coding requirement:** Business users can create applications without extensive programming knowledge.
- **Automation support:** Power Automate allows applications to be connected with automated business workflows.

Limitations

- **Platform dependency:** Organizations can become highly dependent on the Microsoft ecosystem.
- **Customization limits:** Highly specialized applications may require additional coding.
- **Licensing:** Advanced features and enterprise usage can involve additional licensing costs.
- **Complexity:** Large applications may require professional developers for architecture and maintenance.

Best Suited For

- Organizations already using Microsoft 365 and other Microsoft cloud services.
 - Small and medium-sized businesses requiring quick internal applications.
 - Enterprises that need workflow automation and data-driven business applications.
 - Citizen developers and professional developers working together.
-

2. Mendix

Introduction

Mendix is an enterprise low-code application development platform originally developed by Mendix and now part of Siemens. It uses visual modeling to allow teams to develop applications with significantly less traditional programming. The platform supports collaboration between business users, developers, and IT teams during application development. It is designed for organizations developing scalable applications as part of digital transformation initiatives.

Developer

- **Developer:** Mendix, now part of Siemens.
- **Platform type:** Enterprise low-code application platform.
- **Primary focus:** Enterprise application development and digital transformation.
- **Target users:** Professional developers, business users, and enterprise IT teams.

Key Features

- **Visual modeling:** Applications, data models, interfaces, and workflows can be designed visually.
- **Collaborative development:** Business and technical teams can work together on the same application.
- **Enterprise deployment:** Supports cloud-based and enterprise deployment requirements.
- **AI capabilities:** Modern Mendix development incorporates AI-assisted and agentic development capabilities.

Advantages

- **Enterprise scalability:** Suitable for applications used by large organizations.
- **Visual development:** Reduces the amount of traditional programming required.
- **Collaboration:** Helps business and IT teams communicate through visual models.
- **Extensibility:** Developers can add custom code when low-code components are insufficient.

Limitations

- **Cost:** Enterprise-level deployment can require significant investment.
- **Learning curve:** Understanding the complete platform can take time for beginners.
- **Platform dependency:** Applications depend on Mendix's development environment and ecosystem.
- **Complex projects:** Advanced applications may still require experienced developers.

Best Suited For

- Large organizations undergoing digital transformation.
 - Manufacturing and industrial organizations.
 - Enterprises requiring scalable business applications.
 - Teams combining professional developers with business-domain experts.
-

3. OutSystems

Introduction

OutSystems is an enterprise low-code application development platform designed for building web, mobile, and business applications. It combines visual development with AI-assisted development and professional development capabilities. The platform focuses on rapid application delivery while maintaining scalability, security, and governance. It is especially aimed at organizations developing mission-critical and enterprise applications.

Developer

- **Developer:** OutSystems.
- **Platform type:** Enterprise low-code and AI-assisted development platform.
- **Primary focus:** High-performance enterprise applications.
- **Target users:** Professional developers, enterprise IT teams, and business technologists.

Key Features

- **Visual development:** Interfaces, data models, and application logic can be created visually.
- **AI-assisted development:** AI tools assist developers in creating and extending applications.
- **Flexible deployment:** Supports public cloud, regulated or sovereign cloud, and on-premises environments.
- **Enterprise governance:** Provides security, lifecycle management, and governance capabilities.

Advantages

- **High scalability:** Suitable for large and complex applications.
- **Rapid development:** Visual development reduces manual programming effort.
- **Strong enterprise capabilities:** Provides security and governance features.
- **AI integration:** Modern versions combine low-code development with AI-driven development.

Limitations

- **Cost:** Enterprise-oriented licensing can be expensive.
- **Learning curve:** Professional use requires understanding of the platform's development concepts.
- **Platform dependency:** Applications rely on the OutSystems ecosystem.
- **Complexity:** Simple applications may not require the full capabilities of the platform.

Best Suited For

- Large enterprises developing mission-critical applications.
 - Organizations modernizing legacy applications.
 - Financial, healthcare, and government organizations.
 - Professional development teams requiring low-code with advanced customization.
-

4. Appian

Introduction

Appian is a low-code platform that focuses strongly on business process automation and process orchestration. It allows organizations to build applications while connecting people, data, systems, and automated processes. The platform combines low-code development with workflow automation, artificial intelligence, and data management. It is particularly useful for organizations with complex and regulated business processes.

Developer

- **Developer:** Appian Corporation.
- **Platform type:** Enterprise low-code and process automation platform.
- **Primary focus:** Business process automation and orchestration.
- **Target users:** Enterprises, government organizations, and professional developers.

Key Features

- **Process modeling:** Business processes can be visually designed and automated.
- **Workflow automation:** Automates repetitive and multi-step business processes.
- **AI capabilities:** Supports generative AI, AI agents, and AI-assisted application development.
- **Data fabric:** Connects and provides access to data from different enterprise systems.

Advantages

- **Strong workflow automation:** Excellent for complex business processes.
- **Enterprise support:** Designed for large and regulated organizations.
- **AI integration:** Supports AI-based process automation and intelligent applications.
- **Governance:** Provides controlled development and deployment capabilities.

Limitations

- **Cost:** Mainly targeted toward enterprise customers.
- **Learning requirement:** Complex workflow concepts require training.
- **Overkill for simple apps:** Small applications may not need its extensive capabilities.
- **Platform dependency:** Organizations become dependent on Appian's ecosystem.

Best Suited For

- Large enterprises with complex workflows.
 - Banking and financial institutions.
 - Insurance and government organizations.
 - Organizations requiring process automation and case management.
-

5. Zoho Creator

Introduction

Zoho Creator is a cloud-based low-code application development platform developed by Zoho Corporation. It allows users to create custom web and mobile applications using visual development tools and minimal programming. The platform provides application development, workflow automation, analytics, integration, and AI capabilities in one environment. It is used by businesses ranging from small organizations to large enterprises.

Developer

- **Developer:** Zoho Corporation.
- **Platform type:** Cloud-based low-code application platform.
- **Primary focus:** Custom business applications and process automation.
- **Target users:** Small businesses, enterprises, developers, and citizen developers.

Key Features

- **Drag-and-drop development:** Users can design applications without extensive programming.
- **Workflow automation:** Processes can be automated using visual workflows and Blueprints.
- **Integration:** Provides 1,000+ integrations through its ecosystem and connectors.
- **AI capabilities:** AI-assisted application creation, analytics, and workflow features are available.

Advantages

- **Easy to use:** Suitable for beginners and business users.
- **Fast development:** Drag-and-drop tools reduce application development time.
- **Strong integration:** Works naturally with Zoho applications and external services.
- **Multi-device support:** Applications can be used across web, Android, and iOS devices.

Limitations

- **Platform dependency:** Heavy use of Zoho services can create vendor dependence.
- **Advanced customization:** Complex requirements may require Deluge scripting or developer support.
- **Pricing:** Advanced capabilities may require paid plans.
- **Complex enterprise needs:** Extremely specialized applications may be better suited to enterprise-focused platforms.

Best Suited For

- Small and medium-sized businesses.
 - Organizations already using Zoho products.
 - Departments requiring custom internal applications.
 - Citizen developers and teams seeking quick business solutions.
-

6. Salesforce Lightning Platform

Introduction

Salesforce Lightning Platform is a low-code application development environment within the Salesforce ecosystem. It enables organizations to create applications, customize business processes, and automate workflows with minimal coding. The platform is strongly connected to Salesforce's customer relationship management and data capabilities. It supports both non-technical users and professional developers through a combination of low-code and pro-code tools.

Developer

- **Developer:** Salesforce.
- **Platform type:** Cloud-based enterprise low-code platform.
- **Primary focus:** CRM-connected applications and business automation.
- **Target users:** Salesforce administrators, business users, and developers.

Key Features

- **Lightning App Builder:** Enables visual creation of application interfaces.
- **Workflow automation:** Salesforce Flow allows organizations to automate business processes.
- **CRM integration:** Applications can directly use Salesforce customer and business data.
- **AI capabilities:** Low-code tools can be combined with Salesforce's AI and Agentforce capabilities.

Advantages

- **Strong CRM integration:** Works directly with Salesforce customer data.
- **Low-code and pro-code:** Supports both simple configuration and advanced development.
- **Enterprise scalability:** Designed for large organizations and business applications.
- **AI ecosystem:** Provides access to Salesforce's growing AI capabilities.

Limitations

- **Salesforce dependency:** It is most valuable when an organization already uses Salesforce.
- **Cost:** Enterprise Salesforce solutions can be expensive.
- **Learning curve:** Understanding the wider Salesforce ecosystem requires training.
- **Customization complexity:** Advanced applications may require Apex and other Salesforce technologies.

Best Suited For

- Organizations already using Salesforce CRM.
 - Sales and marketing departments.
 - Customer service organizations.
 - Enterprises requiring CRM-connected business applications.
-

7. Google AppSheet

Introduction

Google AppSheet is a no-code/low-code application development platform provided by Google. It allows users to create applications using data from sources such as Google Sheets, databases, and other connected services. The platform emphasizes simple application creation, workflow automation, and integration with Google Workspace. It is particularly useful for organizations that need lightweight business applications without extensive programming.

Developer

- **Developer:** Google.
- **Platform type:** No-code/low-code application platform.
- **Primary focus:** Data-driven business applications and automation.
- **Target users:** Business users, citizen developers, and small development teams.

Key Features

- **No-code application creation:** Applications can be created without traditional programming.
- **Data integration:** Supports spreadsheets, databases, and other data sources.
- **Automation:** Bots, processes, events, and tasks can automate business activities.
- **Google ecosystem:** Integrates naturally with Google Workspace services.

Advantages

- **Easy to learn:** Suitable for users without programming experience.
- **Google integration:** Works effectively with Google Workspace.
- **Rapid development:** Simple applications can be created quickly.
- **Automation:** Provides built-in tools for business process automation.

Limitations

- **Customization:** Complex applications may have limitations compared with traditional development.
- **Platform dependency:** Applications depend on Google's AppSheet ecosystem.
- **Advanced requirements:** Highly specialized applications may require other development approaches.
- **Data complexity:** Applications involving very complex enterprise data may require more advanced platforms.

Best Suited For

- Google Workspace users.
 - Small businesses and departments.
 - Field-service and data-collection applications.
 - Users who need quick applications without traditional programming.
-

8. Oracle APEX

Introduction

Oracle APEX, or **Application Express**, is Oracle's low-code platform for developing secure and scalable web applications. It allows developers to build applications through a browser-based environment using declarative development techniques. APEX has direct access to Oracle Database, making it particularly effective for database-driven applications. It combines low-code development with the ability to add custom code when specialized functionality is required.

Developer

- **Developer:** Oracle Corporation.
- **Platform type:** Database-centered low-code development platform.
- **Primary focus:** Secure enterprise web applications.
- **Target users:** Oracle developers, database professionals, and enterprises.

Key Features

- **Browser-based development:** Applications can be developed and managed through a web interface.
- **Direct database access:** Applications can work directly with Oracle Database.
- **Declarative development:** Many application components can be configured without extensive coding.
- **Responsive interfaces:** Provides components for developing modern web applications.

Advantages

- **Strong database integration:** Excellent integration with Oracle Database.
- **Rapid development:** Declarative tools reduce development effort.
- **Security:** Provides enterprise-oriented security capabilities.
- **Scalability:** Suitable for applications ranging from departmental tools to enterprise systems.

Limitations

- **Oracle dependency:** It is particularly dependent on the Oracle ecosystem.
- **Database knowledge:** Understanding Oracle Database improves effective use of the platform.
- **Web-focused:** It is primarily designed around web applications.
- **Specialized ecosystem:** Organizations using other database technologies may find alternatives more suitable.

Best Suited For

- Organizations using Oracle Database.
 - Database-driven enterprise applications.
 - Internal business applications and dashboards.
 - Developers and database administrators working with Oracle technologies.
-

9. SAP Build Apps / SAP Build

Important for your report: SAP Build Apps is now a **deprecated standalone product**, effective **March 23, 2026**. SAP's current direction is the unified **SAP Build** offering, which combines low-code, pro-code, automation, digital workspaces, and generative AI.

Introduction

SAP Build is SAP's unified platform for developing applications, automating processes, and creating digital workspaces. It combines low-code, pro-code, and generative AI capabilities within the SAP ecosystem. The platform is designed to help business users and professional developers extend SAP applications and business processes. SAP Build is particularly useful for organizations using SAP S/4HANA, SAP BTP, and other SAP business solutions.

Developer

- **Developer:** SAP SE.
- **Current platform:** SAP Build.
- **Earlier platform:** SAP Build Apps, formerly known as AppGyver.
- **Primary focus:** SAP application development, automation, and extensions.

Key Features

- **Low-code development:** Provides visual tools for creating business applications.
- **Process automation:** Supports workflow automation using low-code and no-code tools.
- **AI assistance:** Joule provides AI assistance for application and automation development.
- **Pro-code support:** Supports ABAP, Java, JavaScript/TypeScript, and SAP Cloud Application Programming Model.

Advantages

- **SAP integration:** Provides strong integration with SAP business systems.
- **Unified development:** Combines low-code, pro-code, automation, and AI capabilities.
- **Enterprise governance:** Supports controlled and upgrade-safe SAP extensions.
- **AI support:** Natural-language and AI-assisted development capabilities improve productivity.

Limitations

- **SAP dependency:** It provides the greatest value within the SAP ecosystem.
- **Enterprise complexity:** Understanding SAP environments can be challenging for beginners.
- **Cost:** Enterprise SAP solutions can involve substantial licensing and infrastructure costs.

- **Learning requirement:** Professional use may require knowledge of SAP technologies and business processes.

Best Suited For

- Organizations already using SAP enterprise solutions.
 - Large organizations extending SAP S/4HANA applications.
 - HR, procurement, finance, and supply-chain departments.
 - Enterprises requiring governed low-code and process automation.
-

10. ServiceNow App Engine

Introduction

ServiceNow App Engine is a low-code platform for building custom workflow applications on the ServiceNow platform. It allows organizations to digitize business processes and create applications using visual development tools. The platform combines application development with ServiceNow's workflow, data, security, and automation capabilities. Its latest capabilities also incorporate AI-assisted application development and AI agents.

Developer

- **Developer:** ServiceNow.
- **Platform:** ServiceNow AI Platform.
- **Primary focus:** Workflow-driven enterprise applications.
- **Target users:** Business technologists, administrators, and professional developers.

Key Features

- **Low-code application development:** Applications can be created using visual development tools.
- **Workflow automation:** Business processes can be transformed into digital workflows.
- **AI-assisted development:** Build Agent can generate application components from natural-language requirements.
- **Governance and security:** Security, audit trails, and change management are integrated into the platform.

Advantages

- **Strong workflow capabilities:** Excellent for process-driven applications.
- **Enterprise security:** Security and governance are built into the platform.
- **AI integration:** Supports AI-assisted development and intelligent application features.
- **ServiceNow integration:** Works naturally with existing ServiceNow workflows and data.

Limitations

- **ServiceNow dependency:** The platform is strongly tied to the ServiceNow ecosystem.
- **Cost:** Enterprise ServiceNow solutions can be expensive.
- **Learning curve:** Understanding ServiceNow's platform requires training.
- **Best for workflows:** It may be less suitable for applications unrelated to business process management.

Best Suited For

- Organizations already using ServiceNow.
- IT service management departments.
- HR and employee workflow applications.
- Enterprises requiring governed workflow automation.

Overall Observation from the 10 Platforms :

After studying all ten platforms, an important point becomes clear: **there is no single "best" low-code platform**. Each platform has developed around a particular ecosystem or business requirement.

Key Findings from the Survey

Finding	Observation
Most beginner-friendly	Google AppSheet, Zoho Creator, Power Apps
Strongest enterprise focus	Mendix, OutSystems, Appian, SAP Build, ServiceNow
Strongest workflow focus	Appian, ServiceNow
Strongest CRM focus	Salesforce Lightning
Strongest database focus	Oracle APEX
Strongest Microsoft integration	Power Apps
Strongest Google integration	AppSheet
Strongest SAP integration	SAP Build
Strong AI direction	Salesforce, ServiceNow, SAP Build, Power Apps, Zoho Creator, AppSheet
Best for rapid/simple apps	AppSheet, Zoho Creator, Power Apps
Best for complex enterprise apps	Mendix, OutSystems, Appian, SAP Build, ServiceNow



Comparative Survey of Low-Code Platforms ✨

#	Platform	Development Approach	Ease of Use	Customization	Application Type	Workflow Automation	AI Capabilities	Integration	Scalability	Deployment	Target Users	Best Use Case	Major Limitation
1	Microsoft Power Apps	Low-code + some no-code	Easy	High	Web, Mobile, Enterprise	Advanced	Advanced	400+ data sources; Microsoft + third-party	High	Cloud + On-premise data connectivity	Citizen + Professional developers	Microsoft-based business applications	Microsoft ecosystem dependency
2	Mendix	Low-code	Moderate	Extensive	Web, Mobile, Enterprise	Advanced	Advanced	Enterprise APIs, databases, cloud services	Very High	Cloud, Private Cloud, On-premise	Business + Professional developers	Enterprise digital transformation	Higher learning curve and cost
3	OutSystems	Low-code	Moderate	Extensive	Web, Mobile, Enterprise	Advanced	Advanced	APIs, databases, enterprise systems	Very High	Cloud + On-premise/ Private environments	Professional developers + Enterprises	Complex enterprise applications	Cost and platform complexity
4	Appian	Low-code	Moderate	High	Web, Mobile, Enterprise	Very Advanced	Advanced	APIs, data fabric, enterprise systems	Very High	Cloud + Enterprise deployment options	Professional + Business users	Process automation and case management	Less suitable for simple applications
5	Zoho Creator	Low-code	Easy	High	Web, Mobile, Business	Advanced	Advanced	1,000+ integrations; APIs	High	Cloud, On-premise, Hybrid	Citizen + Professional developers	SMB business applications	Advanced customization may require scripting
6	Salesforce Lightning Platform	Low-code + pro-code	Moderate	Extensive	Web, Mobile, Enterprise	Advanced	Very Advanced	Salesforce ecosystem, APIs, MuleSoft	Very High	Primarily Cloud	Citizen + Professional developers	CRM and customer applications	Strong Salesforce dependency
7	Google AppSheet	No-code	Very Easy	Moderate	Web, Mobile	Advanced	Advanced	Google Workspace, databases, APIs, third-party services	High	Cloud	Mainly Citizen developers	Simple data-driven business apps	Limited customization for complex apps
8	Oracle APEX	Low-code	Moderate	Extensive	Mainly Web, Enterprise	Advanced	Moderate-Advanced	Oracle Database, SQL, REST APIs	Very High	Cloud + On-premise	Professional developers + DB experts	Database-driven applications	Strong Oracle Database dependency
9	SAP Build	Low-code + pro-code	Moderate	Extensive	Web, Mobile, Enterprise	Advanced	Very Advanced	SAP + external databases/APIs	Very High	Mainly Cloud / SAP BTP	Business + Professional developers	SAP business extensions and automation	Best value within SAP ecosystem
10	ServiceNow App Engine	Low-code + no-code + pro-code	Moderate	High	Web, Mobile, Enterprise	Very Advanced	Very Advanced	ServiceNow + enterprise integrations	Very High	Cloud	Citizen + Professional developers	Workflow and enterprise service applications	ServiceNow dependency

Conclusion :

The survey shows that low-code platforms have evolved beyond simple drag-and-drop application builders.

Modern platforms increasingly combine **application development, workflow automation, AI, data integration, security, and enterprise governance**.

The major difference between platforms is their **ecosystem and primary business focus**, rather than simply their ability to create applications.

Therefore, platform selection should be based on organizational requirements, existing technologies, application complexity, scalability, cost, and long-term objectives.

Survey on Emerging Trends in Low-Code Development

Low-code development has emerged as an important approach to modern software development. Low-code platforms allow applications to be created using visual interfaces, reusable components, and minimal hand-written code. The growing demand for faster application delivery, digital transformation, and reduced development costs has increased the adoption of low-code technologies across organizations.

1. AI-Powered Low-Code Development

One of the major emerging trends is the integration of **Artificial Intelligence (AI) and Generative AI** with low-code platforms. AI can assist developers in generating application components, workflows, database structures, and code from natural-language descriptions. This reduces development time and allows developers to focus more on solving business problems rather than writing repetitive code.

2. Growth of Citizen Development

Low-code platforms are increasingly enabling **non-professional developers**, commonly known as citizen developers, to create applications. Business employees can build simple applications and automate workflows without depending entirely on professional IT teams. This improves collaboration between business departments and developers while reducing the pressure on IT resources.

3. Industry-Specific Applications

Low-code development is moving beyond simple internal applications and is being adopted in industries such as **healthcare, banking, manufacturing, education, and government**. Organizations use low-code platforms to automate business processes, modernize legacy systems, manage data, and develop customized solutions according to industry requirements.

4. Integration and Automation

Another important trend is the integration of low-code platforms with **APIs, cloud services, databases, IoT devices, and enterprise systems**. Low-code tools increasingly provide workflow automation capabilities that allow organizations to connect different systems and automate repetitive tasks. This helps improve operational efficiency and reduce manual work.

5. Security, Governance, and Scalability

As low-code adoption increases, organizations are paying greater attention to **security, governance, scalability, and vendor lock-in**. Uncontrolled citizen development can create security and compliance risks. Therefore, organizations are introducing governance policies, access controls, monitoring, and development standards to ensure that low-code applications remain secure and maintainable.

Conclusion

The future of low-code development is moving toward **AI-assisted development, citizen development, intelligent automation, industry-specific solutions, and stronger enterprise integration**. Although challenges such as security, scalability, and vendor dependency remain, low-code platforms are becoming an important part of modern software development. The combination of low-code and Generative AI is expected to further reduce technical barriers and make application development faster and more accessible.